2313666　殷家兴　自动化　　电拖②

~~3-3~~ 3-1 解:

对~~直~~电动机，有

$P_N = U_N I_N \eta_N$　　$I_N = \frac{P_N}{U_N \cdot \eta_N} = 93.1A$

$T_{2N} = \frac{P_N}{\frac{2\pi n_N}{60}} = \cancel{9.55\frac{P_N}{n_N}}\ 9550\frac{P_N}{n_N} = 108.2\ N\cdot m$

额定输入功率

$P_1 = \frac{P_N}{\eta_N} = 29.5\ kW$

3-3 解:

单叠绕组支路对数 $a=p=3$　~~总导体~~

电动势常数 $\cancel{C_e = \frac{pN}{60a}} = \cancel{\frac{pN}{60a}} =$　$C_e = \frac{pN}{60a} = 6.63$

$\cancel{C_e = \frac{pN}{2a\pi} = 63.34}$

转矩~~系~~常数 $C_t = \cancel{9.55\ 6}\ \frac{pN}{2a\pi} = 63.34$

电枢电动势: $E_a = C_e \Phi n$

当 $n_1 = 1500r/min$ 时　$E_{a1} = C_e \Phi n_1 = 208.8V$

当 $n_2 = 500r/min$ 时　$E_{a2} = C_e \Phi n_2 = 69.6\ V$

若电枢电流为 $I_a = 10A$，电磁转矩 $T = C_t \Phi I_a = 13.3N\cdot m$

3-4 解:

额定输出转矩 $T_{2N} = 9550\frac{P_N}{n_N} = 57.3\ N\cdot m$

空载转矩 $T_0 = \cancel{\frac{P_0}{n_N}} = \frac{P_0}{\Omega} = 9.55\frac{P_0}{n_N} = 3.77N\cdot m$

电磁转矩 $T_N = T_{2N} + T_0 = 61.07\ N\cdot m$

电磁功率 $P_M = P_N + p_0 = \cancel{6.5kW}\ 6.395kW$

~~效率 $\eta_N = \frac{P_N}{P_M}$~~

输入功率 $P_{iN} = P_M + \cancel{P_{ca}}\ \cancel{P_{cu}}\ P_{cua} = 6.895kW$

效率 $\eta_N = \frac{P_N}{P_{iN}} \times 100\% = 87\%$

$\cancel{U_N = E_a + I_a R_a}$　$U_N = E_a + I_a R_a$

输入电流　$I_a = \frac{P_{iN}}{U_N} = 31.34A$　　$E_a = \frac{P_M}{I_a} = 204.1V$

~~电阻 $R_a = \frac{U_N}{I_a} =$~~　　$R_a = \frac{U_N - E_a}{I_a} = 0.507\Omega$